EE6406: Analytic and Ensemble Learning

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2026–1

EE6406: Analytic and Ensemble Learning

- L1: Introduction (ZP)
- L2: Data Preprocessing (ZP)
- L3: Linear Parametric Models (KA)
- L4: Learning Score Functions (KA)
- L5: Over- and Under-determined Analytic Regression (KA)
- L6: Advanced Analytic Classification (KA)
- L7: Analytic Methods for Penalized Learning (KA)
- L8: Performance Evaluation and Statistical Inference (ZP)
- L9: Introduction to Ensemble Learning, Bagging and Boosting (Simon)
- L10: Classisical Ensemble Algorithms – Random Forest, Adaboost, Gradient Boosting (Simon)
- L11: Advanced Boosting Algorithms – XGBoost and LightGBM (Simon)
- L12: Reinforcement Learning with Ensemble Methods (Simon)
- L13: Industrial Applications – End-to-end Ensemble Models (Simon)

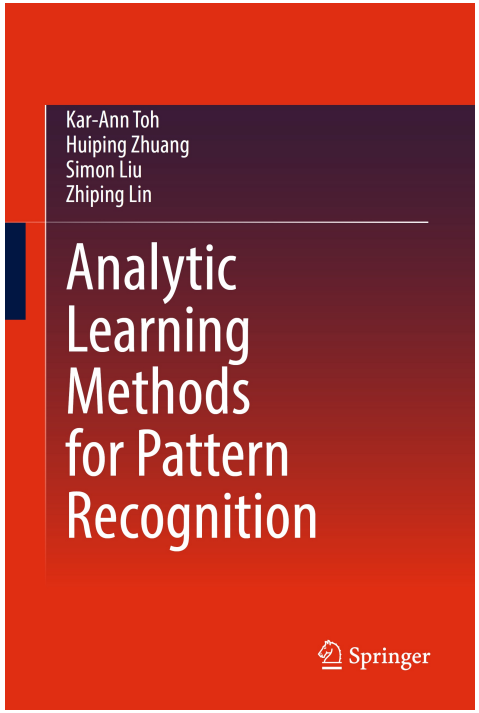
Announcement:

Make-up class for Lunar New Year day:
Zoom recording on 21 Feb., 2026, Sat,
from 9:30am to 12:20pm
Zoom Meeting Link:
https://ntu-sg.zoom.us/j/87272699261?pwd=RbZOtilqgCboz4rrVkL8hr9pzPegyx.1
Meeting ID: 872 7269 9261
Passcode: 712179

EE6406: Analytic and Ensemble Learning

Text Book:

[1] "Analytic Learning Methods for Pattern Recognition" by K.-A. Toh, H. Zhuang, Simon Liu and Z. Lin, Springer, 2025 (https://doi.org/10.1007/978-981-96-2151-4).



This textbook is a consolidation of learning methods which comes in an analytic form. The covered learning methods include classical and advanced solutions to problems of regression, minimum classification error, maximum receiver operating characteristics, bridge regression, ensemble learning and network learning. Both the primal and dual solution forms are discussed for over-and under-determined systems. Such coverage provides an important perspective for handling systems with overwhelming samples or systems with overwhelming parameters. For goal driven classification, the solutions to minimum classification-error, maximum receiver operating characteristics, bridge regression, and ensemble learning represent recent advancements in the literature. In this book, the exercises offer instructors and students practical experience with real-world applications.

References:

[2] "Combining Pattern Classifiers: Methods and Algorithms" (2nd edition) by Ludmila I. Kuncheva, Wiley, ISBN: 978-1-118-31523-1, 2014.

[3] "The Elements of Statistical Learning: Data Mining, Inference and Prediction" (2nd edition) by Trevor Hastie, Robert Tibshirani and Jerome Friedman, Springer, New York, 2017.

[4] Tianqi Chen and Carlos Guestrin, "XGBoost: A scalable tree boosting system," In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785-794, ACM, 2016.

[5] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu, "LightGBM: A highly efficient gradient boosting decision tree," In Advances in Neural Information Processing Systems 30, pp. 3149-3157, NIPS, 2017.

Lecture 1
Introduction

Overview

1. Background
2. Introduction
3. Preprocessing
4. Feature Extraction
5. Learning
6. Performance Evaluation
7. Organization

Breakthrough

https://www.bbc.com/news/technology-35785875

Breakthrough..

https://www.businessinsider.com/googles-alphago-made-artifical-intelligence-history-2016-3

March 2016, AlphaGo – an artificial intelligence program developed by Google DeepMind – played a best-of-five Go match against legendary South Korean grandmaster Lee Sedol in Seoul, South Korea.

Why it was a breakthrough?

- Go is an ancient, extremely complex strategy board game with a vast number of possible positions – far more than chess – making it long believed that computers wouldn't master it soon. (CNBC 2017/05/23)
- Unlike earlier AI like IBM's Deep Blue (which beat the world chess champion in 1997), AlphaGo didn't rely on brute-force calculation but used deep learning and reinforcement learning, enabling it to learn from playing millions of games and to "intuitively" evaluate strong moves. (Forbes 2018/02/07)

Global Impact

- The results shocked the AI and Go communities, because the consensus had been that it would take many more years before AI could defeat a top human Go player. (People's Daily 2016/05/28)
- The match was widely covered by international media and sparked interest in AI research, machine learning, and how these technologies can extend beyond games to real-world problems.

Breakthrough

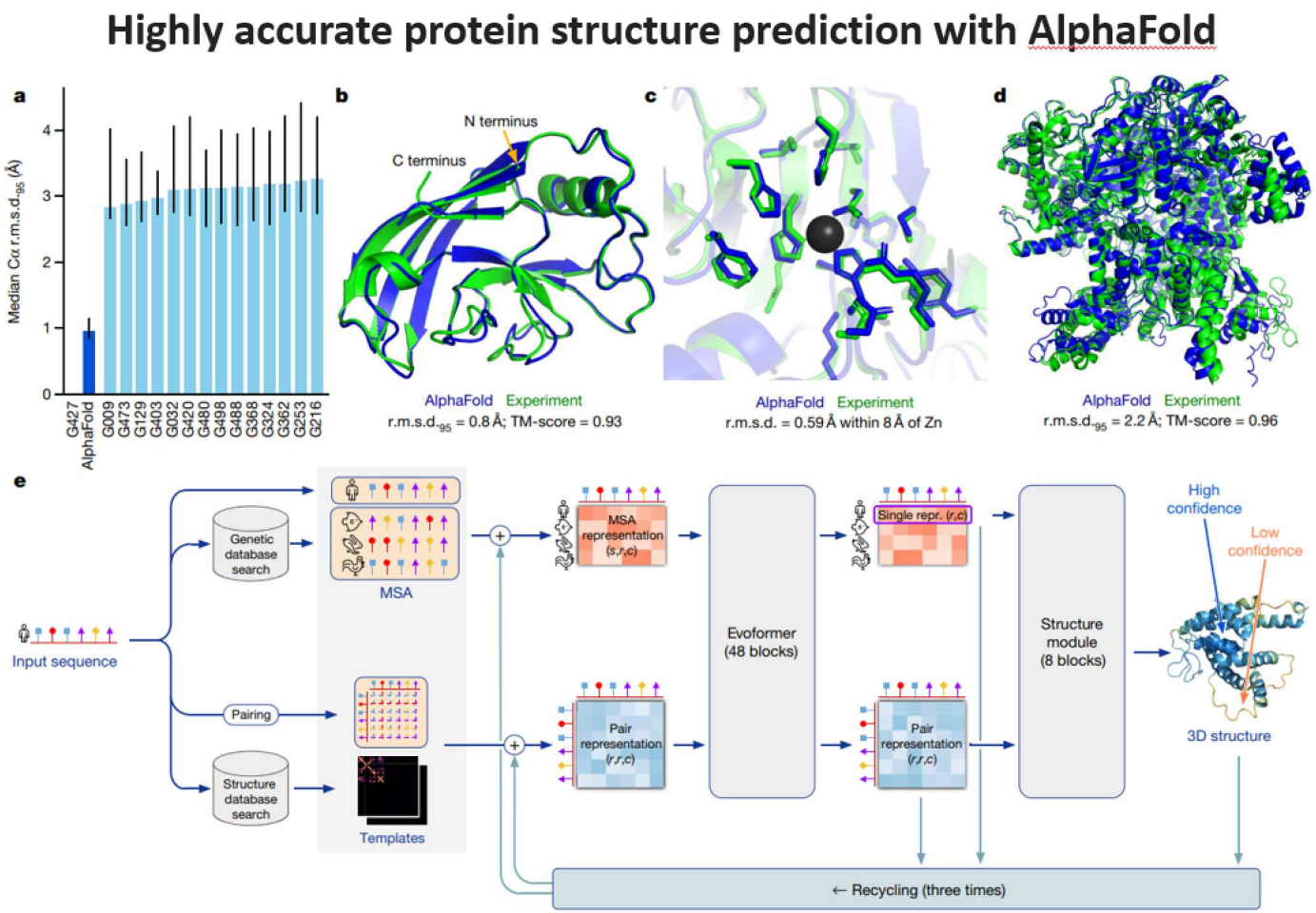


Image credit: https://www.nature.com/articles/s41586-021-03819-2.pdf

Nobel Prize in Chemistry 2024

What the breakthrough was

- AlphaFold is an artificial intelligence system developed by DeepMind that can predict a protein's 3D structure directly from its amino acid sequence with very high accuracy. On average, DeepMind's AI system successfully predicted proteins' three-dimensional shapes to within the width of about one atom. The CASP organizers themselves declared that the protein folding problem had been solved. (Forbes 2021/12/10)
- In 2020 at the CASP14 competition (Critical Assessment of Protein Structure Prediction), AlphaFold dramatically outperformed all other methods and reached levels of accuracy comparable to laboratory experiments– so precise that the organizers said it effectively solved the protein-folding problem. (Nature 2023/03/14)

Breakthrough

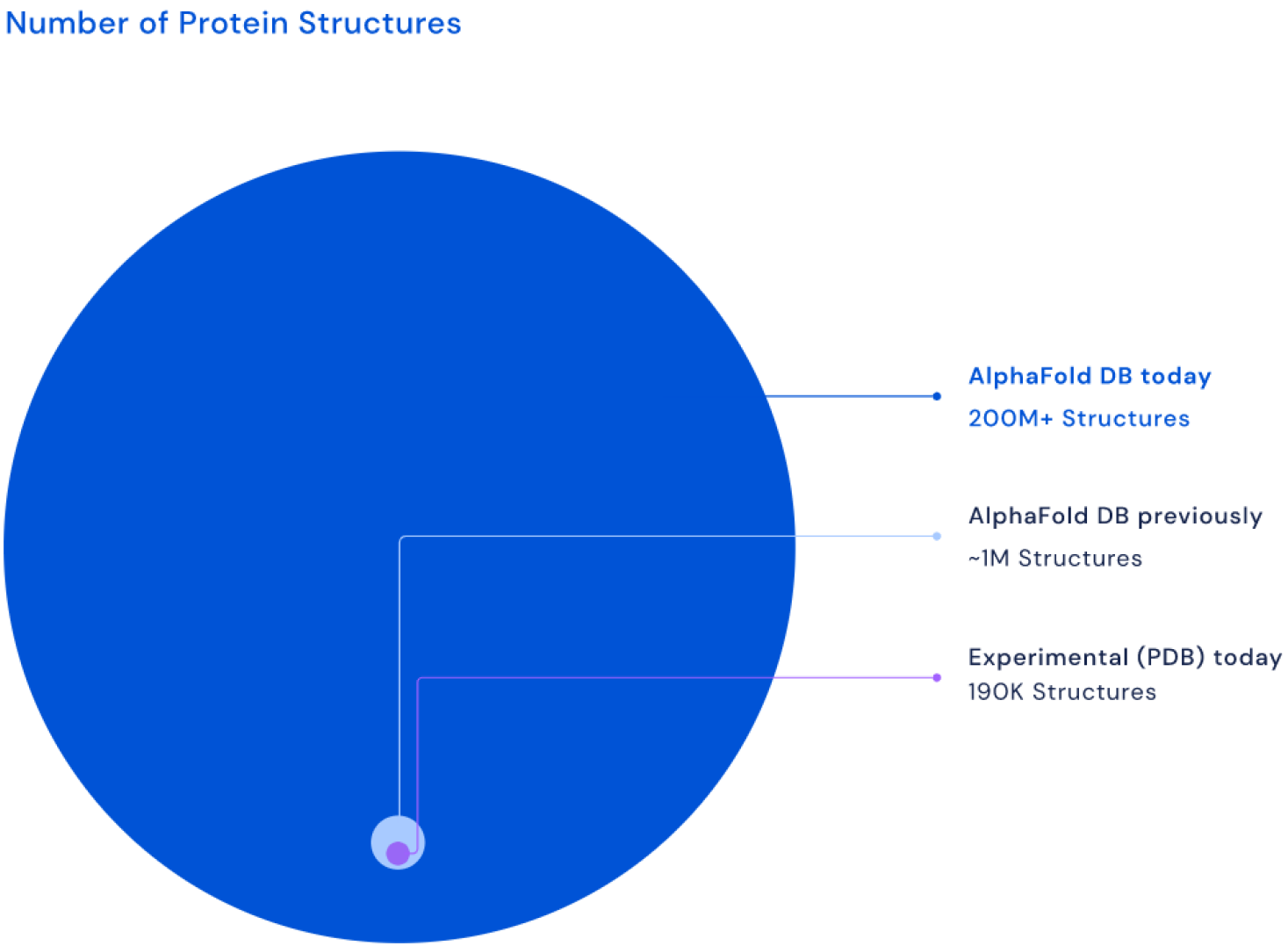


Image source (2022): https://www.deepmind.com/blog/alphafold-reveals-the-structure-of-the-protein-universe

Why it matters

- Protein structure is central to biology: A protein's function depends on its 3D shape – which historically could only be found through slow, costly experimental methods like X-ray crystallography or cryo-EM. (Google DeepMind)
- Huge leap in accuracy and speed: AlphaFold's predictions are often within the width of a single atom of the real structure – a level orders of magnitude better than previous computational methods. (Forbes 2021/12/10)

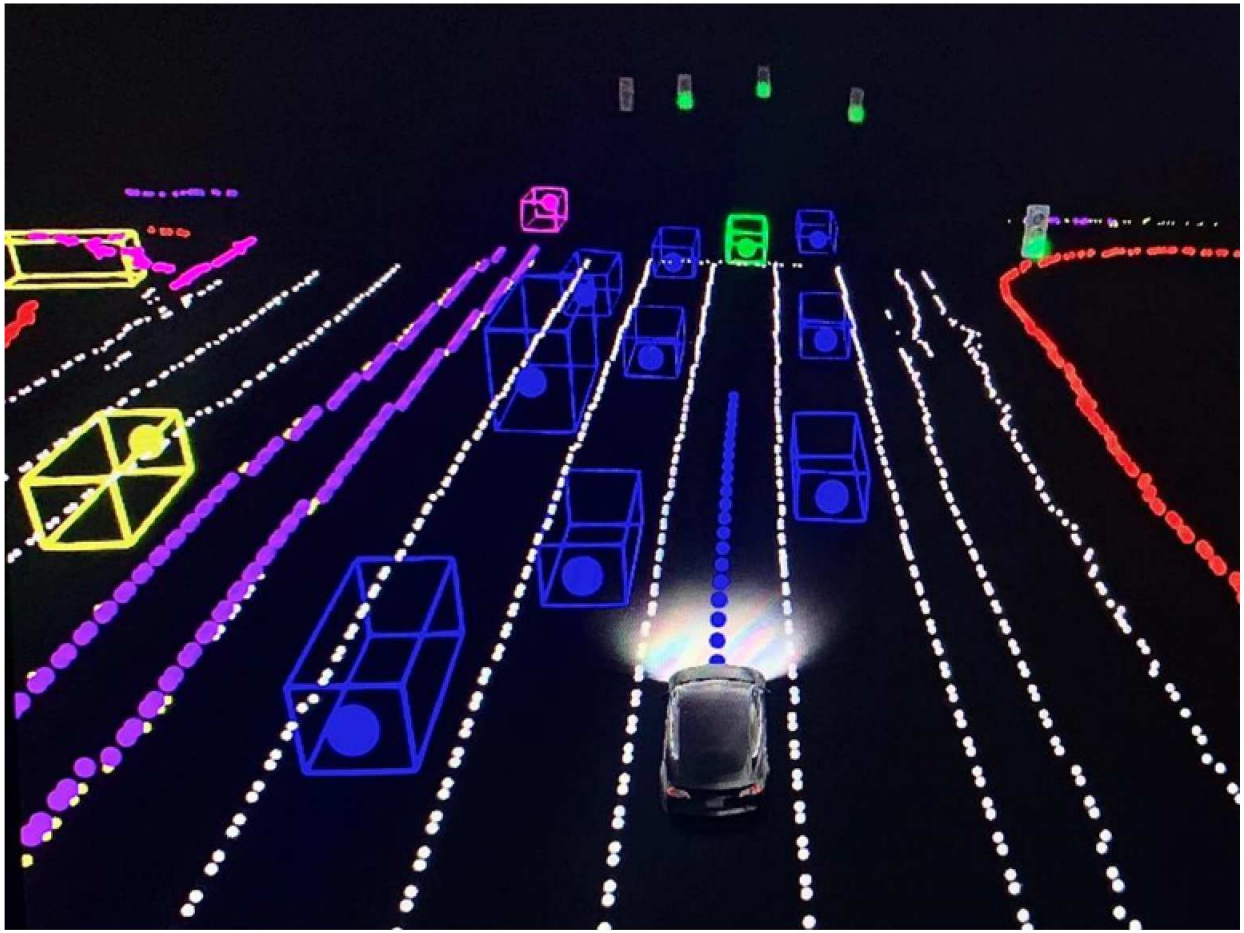
Impact on science and medicine

AlphaFold's success has transformed biomedical research:

- Protein structure prediction is now high-throughput – what once took years experimentally can be done in hours or less. In 2021–22, the AlphaFold Protein Structure Database was released, with hundreds of millions of protein structures freely available worldwide. (Google DeepMind)
- Researchers are using predictions to study disease mechanisms, design better drugs, improve enzyme engineering, and speed biological discovery. (NIH 2024/11/09)

Recently

An image of how Tesla's full self driving (FSD) artificial intelligence sees the world. (TESLA)



https://www.trtworld.com/magazine/five-recent-breakthroughs-in-ai-you-probably-missed-57056

Full Self-Driving (FSD) is Tesla's flagship advanced driver-assistance system designed to enable vehicles to navigate, steer, change lanes, handle intersections, and drive through city and highway environments with minimal human input – moving toward true autonomous driving. It builds on earlier Autopilot systems and is offered as a paid upgrade or subscription. (SCMP 2024/05/09)

Breakthrough Features & Capabilities

SCMP 2024/05/09:

- Advanced navigation: FSD can handle complex highway merges, lane changes, city turns, and stop-controlled intersections with supervised autonomy.
- End-to-end AI: Tesla's neural networks interpret raw image data to directly output steering, braking, and acceleration actions – a paradigm shift from rule-based decision systems.
- Constant fleet learning: Real-world driving data from millions of vehicles continuously feeds model improvements, enabling rapid iteration and refinement.

Recently

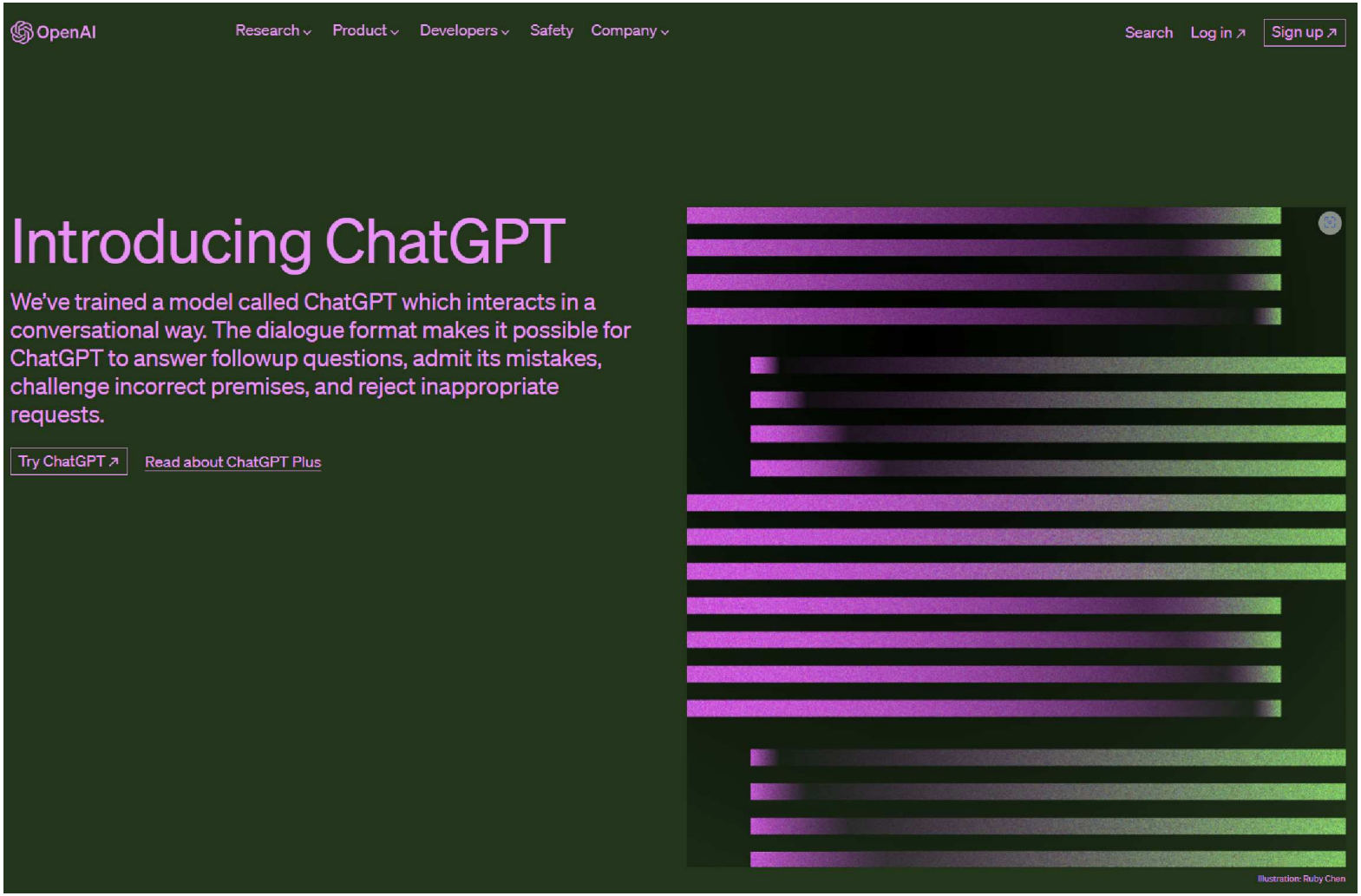


Image source: https://openai.com/index/chatgpt/

ChatGPT was first introduced to the public on November 30, 2022.

Why ChatGPT Was a Turning Point?

Before ChatGPT:

- AI systems were specialized
- Interaction required technical expertise
- Capabilities were fragmented

After ChatGPT:

- AI became conversational, general, and adaptive
- Language became the primary human–AI interface
- AI shifted from a backend tool to a frontline productivity partner

What Made these Happen

- Massive amount of data
- Hardware: CPU, GPU and TPU
- Software: Open Source Platform and Ecosystem
- Algorithms: Advanced Algorithms and Model Architectures
- Feedback Loops and Continuous Learning

Massive Amounts of Data

Why it matters: Data is the raw material of learning.

- Large, diverse datasets allow models to generalize rather than memorize.
- Real-world data (images, text, protein sequences, driving logs) captures complexity that handcrafted rules cannot.
- Self-play and simulation (e.g., AlphaGo) and fleet data (e.g., Tesla) multiply effective data volume.
- Data scale often correlates directly with performance improvements.

Bottom line: Without large-scale data, modern deep learning would not work.

Hardware: CPU, GPU and TPU

Why it matters: Training large models is computationally intensive.

- GPUs enabled massive parallel computation, making deep neural networks practical.
- TPUs and AI accelerators further optimized matrix operations used in deep learning.
- Hardware improvements reduced training time from months to days or hours.
- Energy efficiency gains made large-scale AI economically viable.

Bottom line: Hardware turned theoretical models into deployable systems.

Software: Open Source Platform such and Ecosystem

Why it matters: Innovation scales when tools are shared.

- Frameworks like TensorFlow, PyTorch, CUDA, and scientific libraries lowered barriers to entry.
- Open datasets, benchmarks, and pretrained models accelerated progress.
- Researchers and engineers worldwide could replicate, improve, and combine ideas rapidly.
- Industry-academia feedback loops sped up innovation.

Bottom line: Open ecosystems transformed AI from isolated research into a global movement.

Algorithms: Advanced Algorithms & Model Architectures

Why it matters: Scale alone is not useful without the right learning methods.

- Breakthroughs in deep neural networks, transformers, reinforcement learning, and self-supervised learning unlocked new capabilities.
- End-to-end learning replaced brittle rule-based systems.
- Architectural innovations improved stability, efficiency, and accuracy.
- Algorithms learned representations directly from data instead of relying on human-engineered features.

Bottom line: Algorithmic innovation turned data and compute into intelligence.

Feedback Loops & Continuous Learning

Why it matters: Breakthrough systems improve after deployment.

- Models learn from their own outputs (self-play, simulation).
- Real-world usage feeds continuous retraining.
- Performance gains compound over time rather than plateau.

Bottom line: AI systems now evolve dynamically, not statically.

Introduction

This course is about Analytic and Ensemble Learning for Pattern Recognition.

Motivation

- Presently, courses in the field of machine learning and artificial intelligence place significant emphasis on deep learning.
- Often, such emphasis assumes that students already possess a solid foundation in the classical aspects of machine learning.
- However, essential and potent topics like analytic learning and ensemble learning seemingly receive less attention in most curricula.

Bottom line: Current machine learning education heavily focuses on deep learning, often assuming prior knowledge of classical methods, which leads to important foundational approaches–such as analytic learning and ensemble learning–being underemphasized in most curricula.

Introduction

Motivation

- In particular, analytic learning offers distinct advantages, including determinism, analytical rigor, and interpretability.
- As deep learning systems grow increasingly complex, understanding their behavior becomes more challenging.
- The trend toward intricate models inevitably leads to solutions that defy human intuition and control.

Bottom line: Analytic learning is valuable because it is clear and interpretable, while increasingly complex deep learning models are becoming harder to understand, trust, and control.

- What is Pattern Recognition?
  - Human beings inherently possess the ability to recognize shapes, patterns, and trends.
  - However, replicating this task in a machine is not straightforward. Multiple processing steps are involved in achieving this goal.

Introduction

Pattern Recognition Process

- The initial step is machine perception or image acquisition, where sensors such as cameras or radio receivers come into play. During this stage, the sensed signals are converted into numerical values.
- The subsequent processing step leverages these numerical values for shape or pattern inference. This step constitutes the core of machine learning and artificial intelligence, where meaningful interpretations can be derived from the information processing.
- Finally, the results are utilized for subsequent actions. This entire process is referred to as pattern recognition, which is commonly found in machine learning applications.

Introduction

- What is Analytic Learning?

  https://link.springer.com/referenceworkentry/10.1007/978-1-4419-1428-6_490

  - The word analytic comes from the Greek word 'analytikos', which means having the ability to analyze and divide the whole into its components or elements.
  - Analytic Learning is an analytical approach to learning that uses prior knowledge as a base from which concepts can be described, hypotheses can be developed, and concepts can be rationally generalized by analyzing the components and the structure of the concepts.
  - Analytic learning allows the learner to process information, break it into its component parts, and generate hypotheses by using critical and logical thinking skills.

Introduction

- What is Ensemble Learning?
  - Ensemble learning constructs a composite model from a group of base learners or models often referred to as 'weak learners' when their individual performance is modest. However, when combined, these weak learners form a 'strong learner' capable of achieving superior accuracy.
  - The fundamental premise is that by aggregating predictions from multiple models, the ensemble compensates for individual weaknesses, reduces the risk of overfitting, and manages the bias-variance trade-off more effectively than single models.
  - Ensembles have found applications in finance, where interpretability is crucial. For example, ensemble models like Random Forest or XGBoost offer predictive power similar to deep learning models while maintaining a high level of interpretability, which satisfies financial regulators.

Example 1.1

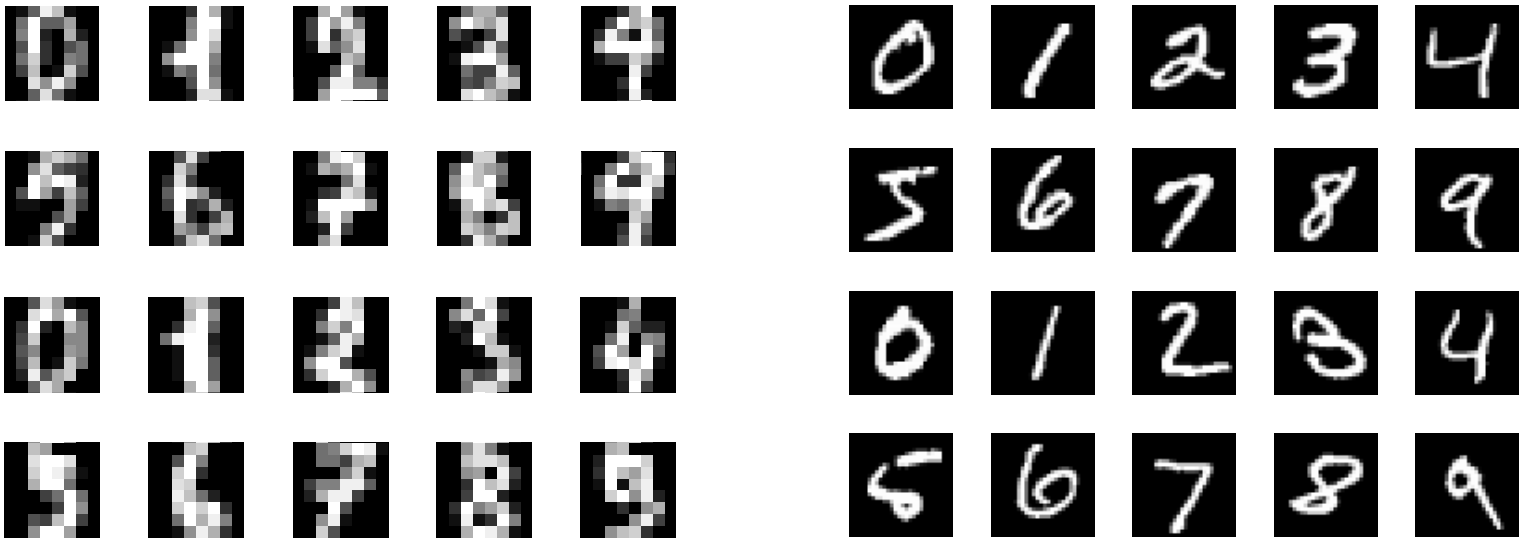


Figure 1: Samples of Optdigit (left panel) and MNIST (right panel) data sets. Samples shown in the upper two rows are taken from the training set and samples shown in the bottom two rows are taken from the test set.

Example 1.1 cont'd

In this example, the machine is to recognize handwritten digits captured as images as shown in Fig. 1.
Here, we showcase two sets of data samples which come with different resolutions. The sample images on the left panel, drawn from the optdigits database [a], are 8×8 pixels grayscale maps, with each pixel ranging in intensity from 0 to 16. The total number of images in this database is 5,620.
The sample images on the right panel, drawn from the MNIST database [b], are 28×28 pixels grayscale maps, with each pixel containing floating-point numbers representing grayscale intensities ranging from 0 (black) to 1 (white). The total number of images in this database is 70,000. Each of these data sets contains handwritten digits numbered from 0 to 9.

[a] https://archive.ics.uci.edu/dataset/80/optical+recognition+of+handwritten+digits
[b] https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits

Example 1.2

Figure 2: Samples of ten identities drawn from the CMU face data set [a]

[a] This dataset was created by Tom Mitchell. The dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license (https://creativecommons.org/licenses/by/4.0/legalcode). The dataset is available at https://archive.ics.uci.edu/dataset/124/cmu+face+images

Example 1.2 cont'd

In this example, the task is to recognize images of faces captured under variations in pose, expression, and the presence of glasses.
Compared with handwritten digits, the degree of variation is larger because it involves imaging three-dimensional objects (faces).
Fig. 2 shows several face image samples drawn from the database of CMU face images.
This database contains 640 face images from 20 subjects with each subject contributing 32 different face images. Each face image of full resolution has 120×128 pixels, with 256 grey levels per pixel. Images of half and quarter resolutions are also available. Face recognition plays a crucial role in various applications, ranging from highly accurate custom and banking systems to more user-friendly mobile device control applications.

Example 1.3

- In the domain of finance, credit risk modeling is a critical tool for assessing the probability that a borrower may default on their loan commitments.
- This evaluation is pivotal in managing the level of risk involved in extending credit, influencing a wide range of financial decisions from personal loans and credit card approvals to broader institutional lending.
- Credit risk modeling aligns with the broader theme of recognizing patterns in data, similar to digit and face recognition examples previously discussed.
- However, instead of identifying digits or facial features, here the focus is on discerning patterns in financial activities that indicate potential default risks.

Example 1.3 (cont'd)

- Traditionally, credit scoring has been predominantly executed using logistic regression.
- This method is favored for its simplicity and the clarity with which it conveys information, making the interpretation and explanation of results straightforward.
- In financial modeling, explainability is crucial as stakeholders, including regulators and loan officers, require clear justifications for decisions that affect financial risk management.

Example 1.3 (cont'd)

- However, with the advancement in data storage capabilities and computational power, financial institutions now harness large volumes and diverse types of historical customer data. This shift has led to the adoption of more sophisticated machine learning techniques.
- Among these, ensemble learning methods like Random Forests and XGBoost have gained prominence.
- These models have consistently outperformed traditional logistic regression in terms of precision and accuracy, providing more reliable assessments of credit risk.

Example 1.3 (cont'd)

- A significant advantage of ensemble methods like Random Forest and XGBoost is their inherent explainability. Despite their complexity, these models facilitate a degree of transparency that allows users to understand the factors driving the predictions. This feature is particularly beneficial in financial applications where the rationale behind a credit scoring decision must be clear and justifiable.
- From Lecture 9 onward, we will delve into these ensemble learning methods, exploring how they enhance predictive performance while maintaining the necessary transparency for financial applications. This exploration will include detailed discussions on the implementation and advantages of using these advanced techniques in credit risk modeling.

Terminologies

Pattern recognition, Machine learning and Data mining

- Pattern recognition: inclined towards engineering and computer vision applications
- Machine learning: more towards statistical and mathematical developments
- Data mining: involves the process of discovering patterns in large data sets under the umbrella of computer science

Terminologies

Classification, Recognition/Identification and Verification

- Classification: refers to either a *binary* decision or *n-ary* decisions
- Recognition and Identification: n-ary decisions
- Verification: only refers to binary decision

A variable is binary or dichotomous when it has only two categories, values or levels. A binary decision is a choice between two alternatives. A dichotomy refers to a partition of a set into two subsets which are jointly and mutually exclusive. For instance, the choice between a 'yes' and a 'no' action is a dichotomous or binary decision.

For a variable or a decision with more than two categories, the terms multi-category, mutli-class, polychotomous can be used interchangeably.

Terminologies

Figure 3: Common Terminologies

Decision Connotation	
Binary	n-ary
Dichotomous Two-category Two-class	Polychotomous Multi-category Mutli-class
Classification Verification –	Classification Identification Recognition

Terminologies

Figure 4: Machine Learning Problems

	Supervised Learning	Unsupervised Learning
Discrete	Classification	Clustering
Continuous	Regression	Dimension Reduction

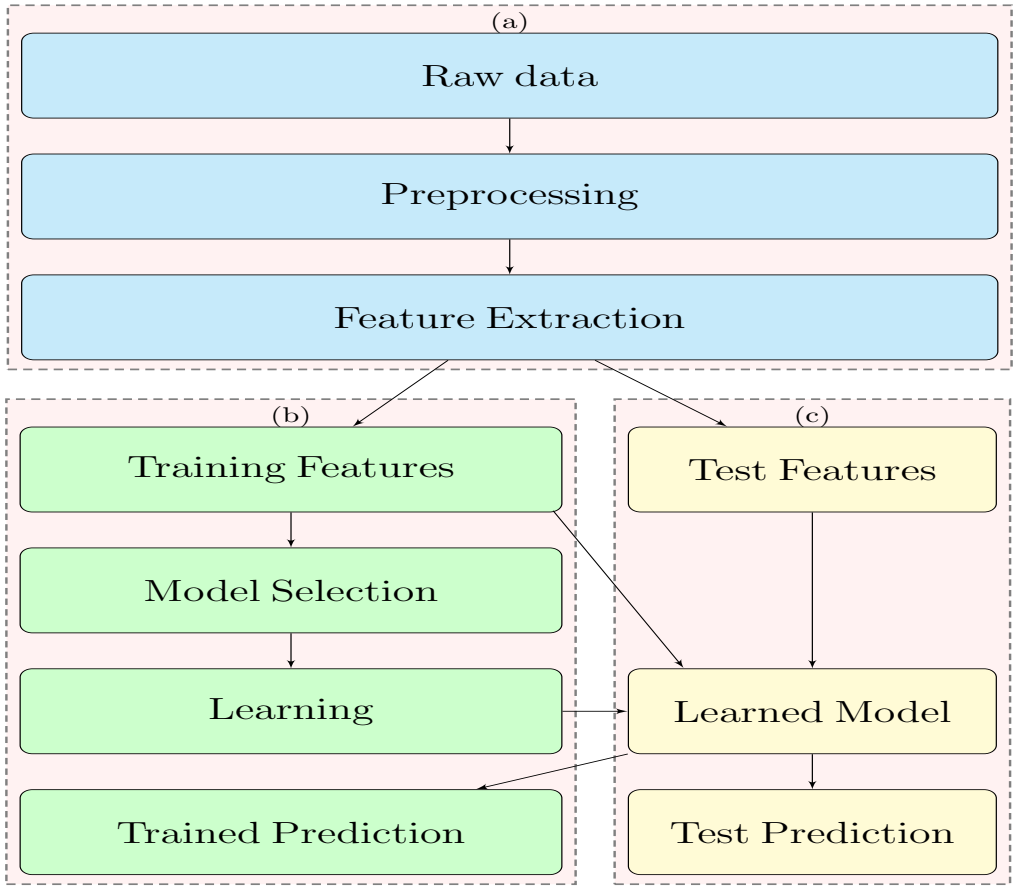
Scope

This lecture focuses on learning based methods for pattern classification with particular focus on analytic or deterministic learning methods.

The complete pipeline for the recognition process will be introduced with emphasis on common building blocks such as learning models, learning cost functions, classifier learning and performance evaluation.

Both the binary and the multi-category problems shall be covered.

Pattern Recognition Process



Data Preprocessing

- Prepares the data for meaningful representation.
- Normalization: the process of removing non-representative or redundant parts of data and make relevant adjustment. The main purpose of data normalization is to prevent anomalies from dominating the analysis.
- Data conversion: conversion among different data types such as *nominal*, *ordinal*, *interval*, and *ratio data* attributes.

Feature Extraction

- *Feature extraction* is the process of extracting informative, relevant and non-redundant components from the bulk of data to facilitate subsequent learning.
- Essentially, it involves a reduction of the amount of resources required for describing a large set of data.
- This process often largely reduces the dimension of data and is thus related to dimension reduction.
  - feature extraction: often involves the process of data transformation.
  - feature selection: also known as variable selection, is the process of selecting a subset of relevant features (variables, predictors) for use in model construction.

Feature Extraction

Approaches to feature extraction can be broadly categorized into the approach of generic *dimension reduction* and the approach of extracting *heuristic* or *semantic features*.

- Dimension reduction: *principal component analysis* (PCA), *independent component analysis* (ICA), *isomap*, *multilinear subspace learning* and *autoencoder*.
- Extracting semantic features: *edge detection*, *corner detection*, *blob detection*, *ridge detection* and *scale-invariant feature transform*.

Learning

Essentially, machine learning in our context can be summarized as learning a function f that maps input variables to output variables i.e.,

$$Y = f(X)$$

where X and Y are respectively generic symbols for input and output variables which can take either scalar or vector quantities.

The form of f, if it exists, is unknown in practice and the job is to evaluate those algorithms at hand that best describe the problem.

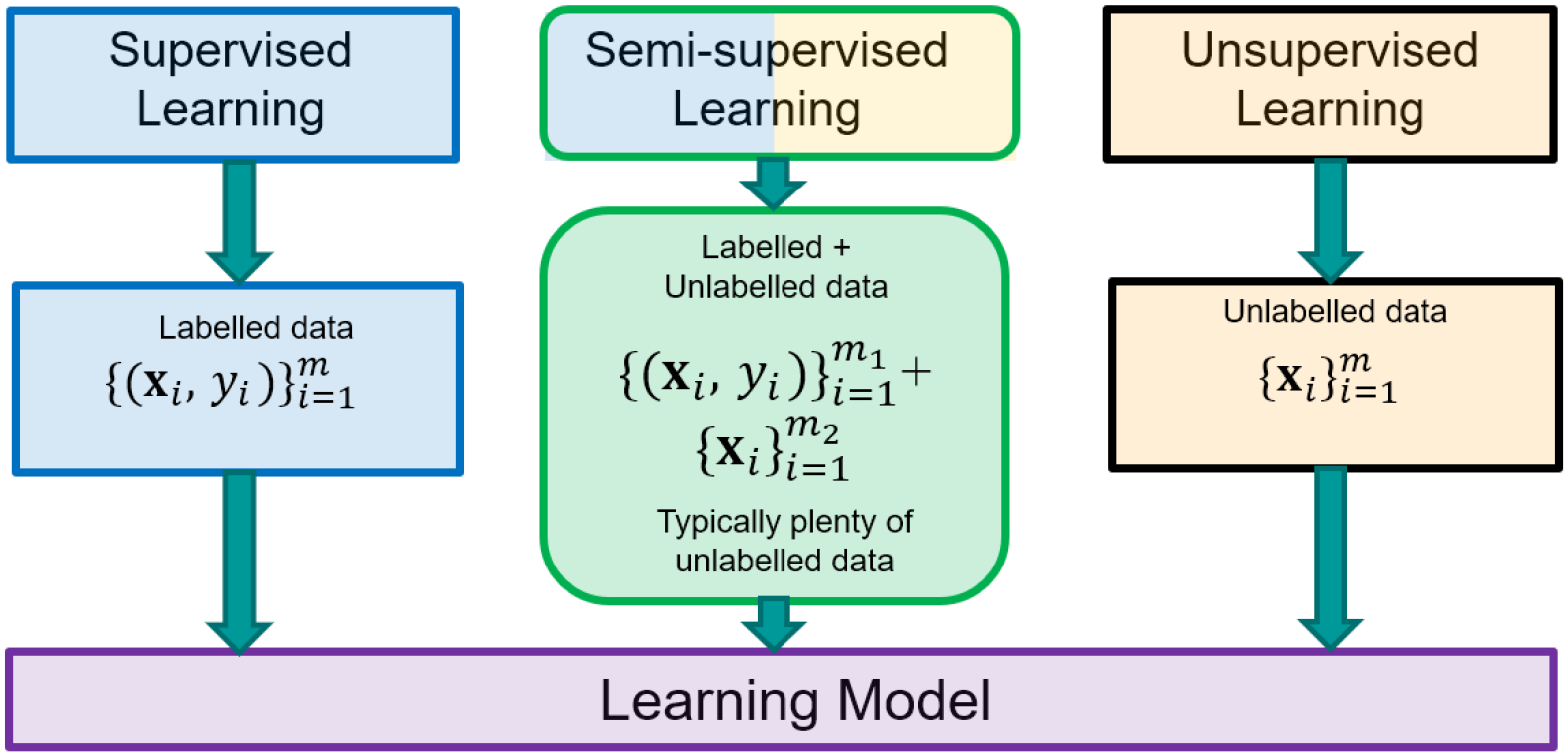
Approaches

- Parametric: assumes a set of parameters of fixed size. E.g., Logistic Regression, Linear Discriminant Analysis, Perceptron, Naive Bayes, and Simple Neural Networks.
- Nonparametric: do not make strong assumptions about the form of the mapping function. E.g., k-Nearest Neighbors, Decision Trees such as CART and C4.5, and Parzen windows.

Approaches

- Supervised Learning: a "teacher" or a target output label is available where the goal is to learn a general rule that maps the inputs to the outputs.
- Unsupervised Learning: no labels are given and the algorithm is to find the structure of the given inputs.
- Semi-supervised Learning: both labeled data and unlabeled data are considered for learning.

Approaches

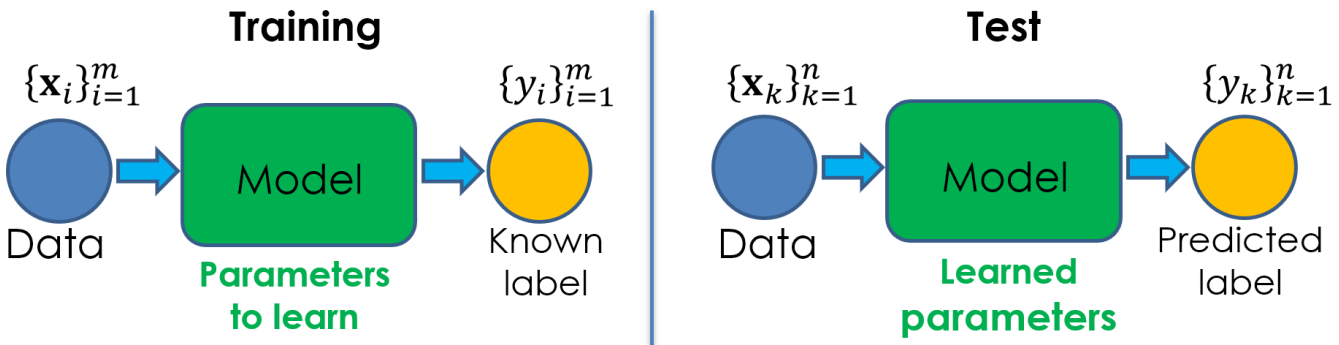

Supervised Learning
Semi-supervised Learning
Unsupervised Learning
Labelled data
$\{(\mathbf{x}_i, y_i)\}_{i=1}^{m}$
Labelled + Unlabelled data
$\{(\mathbf{x}_i, y_i)\}_{i=1}^{m_1} + \{\mathbf{x}_i\}_{i=1}^{m_2}$
Typically plenty of unlabelled data
Unlabelled data
$\{\mathbf{x}_i\}_{i=1}^{m}$
Learning Model

Learning Components

- Data Features: X
- Target Labels: Y
- Learning Model: $f(X, W)$ where W is a set of parameters
- Learning Cost Function: $J(f(X, W), Y)$

Learning Components

- Learning is a mathematical optimization that is performed on a predictive model of fixed structure such as $f(X, W)$.

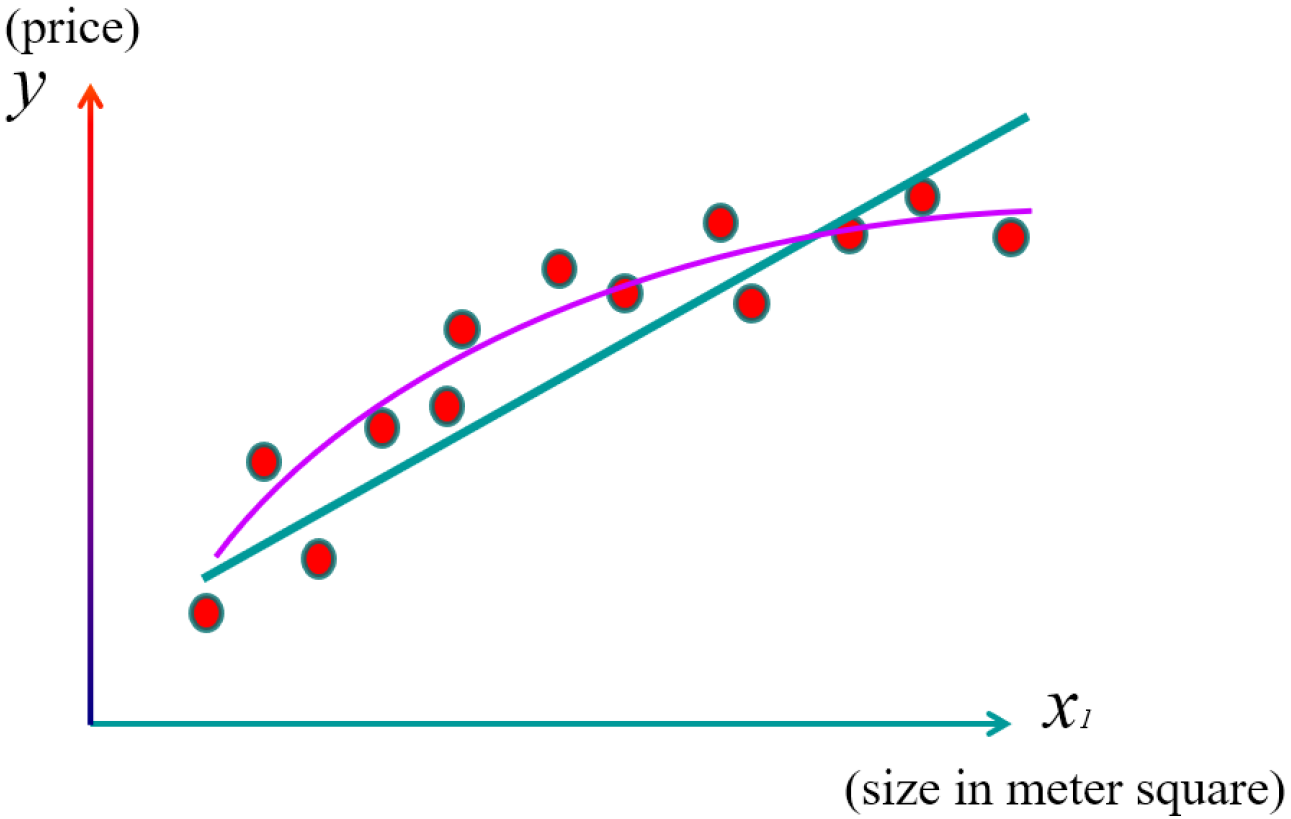


Goal: to learn the model's parameters from the given data and labels $\{\mathbf{x}_i, y_i\}_{i=1}^m$

Goal: to predict the label of novel data using the learned parameters $\{\mathbf{x}_k\}_{k=1}^n$

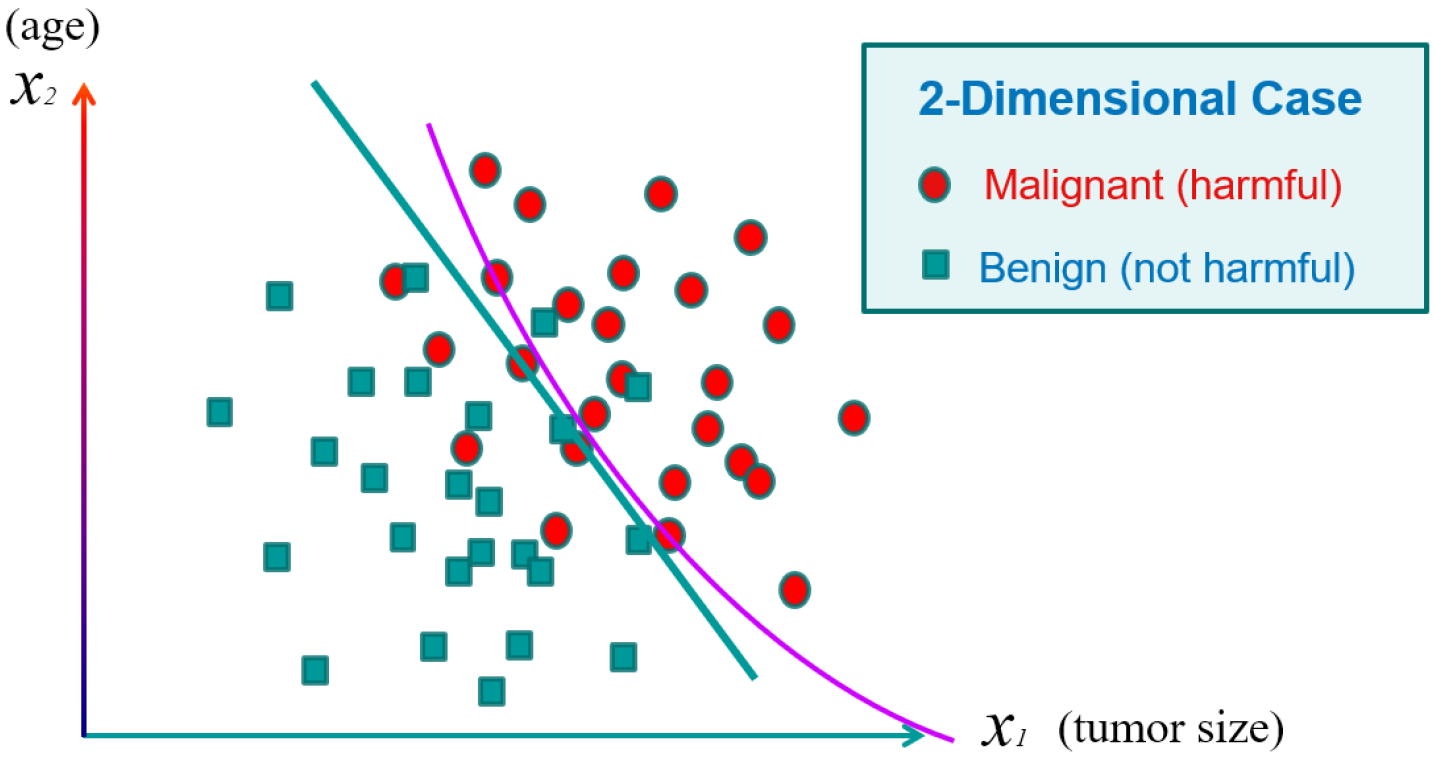
Learning Components

- Regression is a learning process to fit a continuous target ($f(X, W) \longrightarrow Y_{continuous}$). For example, the house price according to the size in meter square can be learning by a model.



Learning Components

- Classification is a categorization decision process upon the output of the learned model ($f(X, W) \longrightarrow Y_{discrete}$).



Learning Components

- For binary classification, when the model output is expressed as a *distance metric* or *confidence level*, the process of categorization boils down to the problem of finding an appropriate decision threshold for data grouping.
- The binary classification can be extended to multi-category classification by adopting a *one-hot* or *one-versus-all* (also known as *winner-takes-all*) decision technique on the multiple binary decisions.

Performance Evaluation

- In real-world applications, the size of sampling is often limited due to acquisition and cost constraints.
- The goodness of learning is thus hinged upon the *predictivity* or *generalization* of the learned model based on the limited size of learning data samples.

Performance Evaluation

- Although the full information regarding the underlying data distribution is not known, the goodness of learning can nevertheless be assessed through an appropriate performance evaluation. This is frequently achieved by isolating a portion out of the entire data set to test the generalization ability of the learning algorithm.
  - To capitalize on all the available data, an N-fold cross-validation test can be performed. This is done by partitioning the entire data set into N portions and then use the $N-1$ portion for learning and the remaining N portion for testing. The process is repeated to test each and every portion of the N portions. The final result is often recorded in averaged form for the N number of tests.

Organization

Lecture 9–13: Ensemble Methods and Applications

Lecture 8: Performance Evaluation and Statistical Inference

Lecture 7: Analytic Methods for Penalized Learning

Lecture 6: Advanced Analytic Classification

Lecture 5: Over- and Under-determined Analytic Regression

$$\hat{\mathbf{W}} = \begin{cases} (\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{Y} & \text{Over-determined} \\ \mathbf{P}^T(\mathbf{P}\mathbf{P}^T + \lambda\mathbf{I})^{-1}\mathbf{Y} & \text{Under-determined} \end{cases}$$

Lecture 4: Learning Cost Function

$$J(\boldsymbol{w}) = \sum_{i=1}^{m} L(f(\boldsymbol{w}, \mathbf{x}_i)) + \lambda R(\boldsymbol{w})$$

Loss Function	Regularization
$L(f(\boldsymbol{w}, \mathbf{x}))$	$\lambda R(\boldsymbol{w})$

Lecture 3: Model	Parameter
$f(\boldsymbol{w}, \mathbf{x})$	$\boldsymbol{w}$

Lecture 2: Data Preprocessing

Lecture 1: Introduction

Thank you!